LAB EXPERIMENT RECORD

Set 18: Product Inventory Management

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Date	10/10/2026
Day	Saturday

AIM

To analyze product inventory data using Tableau by creating a bar chart to visualize product quantities, a stacked bar chart to represent quantities across product categories, a table to display product inventory details, and a dashboard combining all visualizations.

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

plot(employee$Years_of_Service,
     employee$Performance_Score,
     type = "b",
     main = "Employee Performance Trend",
     xlab = "Years of Service",
     ylab = "Performance Score",
product <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

barplot(product$Quantity_Available,
        names.arg = product$Product_Name,
        main = "Quantity of Each Product",
        xlab = "Product Name",
        ylab = "Quantity Available",
        col = "skyblue",
        ylim = c(0, 350))

text(x = seq_along(product$Quantity_Available),
     y = product$Quantity_Available,
     labels = product$Quantity_Available,
     pos = 3)
     pch = 19)
```

RStudio

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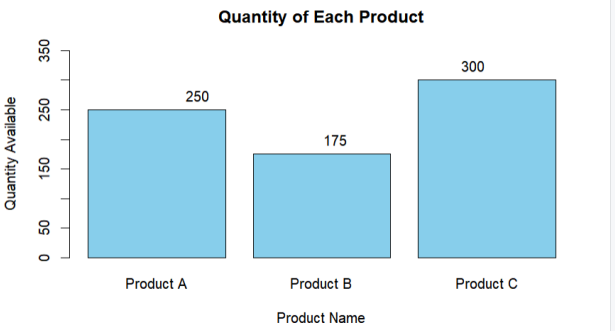
```
product <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

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        main = "Quantity of Each Product",
        xlab = "Product Name",
        ylab = "Quantity Available",
        col = "skyblue",
        ylim = c(0, 350))

text(x = seq_along(product$Quantity_Available),
     y = product$Quantity_Available,
     labels = product$Quantity_Available,
     pos = 3)
```

```
+   Product_Name = c("Product A", "Product B", "Product C"),
+   Quantity_Available = c(250, 175, 300),
+   Price = c(20, 15, 18)
+ )
>
> # Create bar chart
> barplot(product$Quantity_Available,
+         names.arg = product$Product_Name,
+         main = "Quantity of Each Product",
+         xlab = "Product Name",
+         ylab = "Quantity Available",
+         col = "skyblue",
+         ylim = c(0, 350))
>
> # Display values on bars
> text(x = seq_along(product$Quantity_Available),
+      y = product$Quantity_Available,
+      labels = product$Quantity_Available,
+      pos = 3)
>
```

lo	List of 18
long_data	80 obs. of 3 variables
matrix_data	num [1:14, 1:10] 1 1 1 1 1 0 0 0 0 0 ...
model	List of 12
monthly_avg	3 obs. of 2 variables
online_data	6 obs. of 3 variables
pie_data	5 obs. of 5 variables
prec_data	12 obs. of 3 variables
product	3 obs. of 4 variables



29°C Partly sunny

Search

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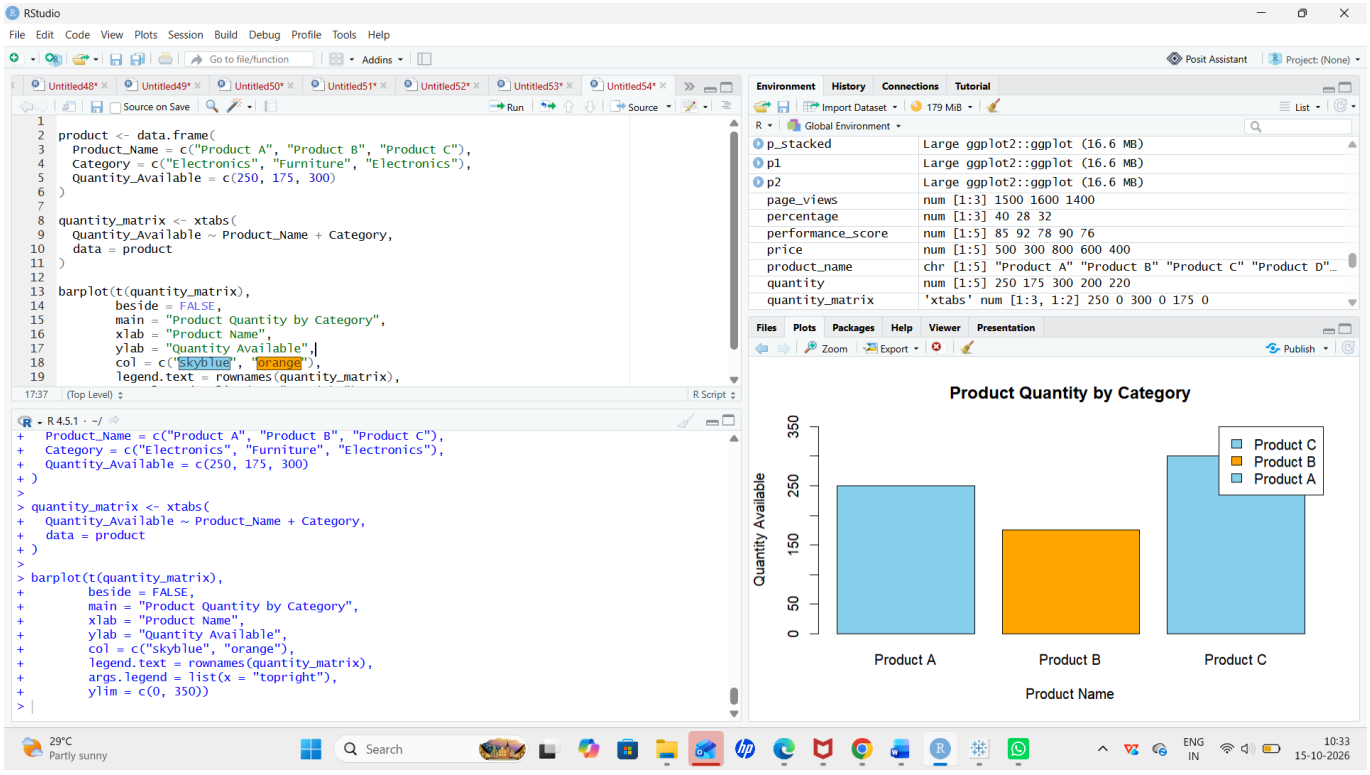
10:31 15-10-2026

2. Generate a bar chart showing the distribution of employees across different departments.

```
product <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C"),
  Category = c("Electronics", "Furniture", "Electronics"),
  Quantity_Available = c(250, 175, 300)
)

quantity_matrix <- xtabs(
  Quantity_Available ~ Product_Name + Category,
  data = product
)

barplot(t(quantity_matrix),
        beside = FALSE,
        main = "Product Quantity by Category",
        xlab = "Product Name",
        ylab = "Quantity Available",
        col = c("skyblue", "orange"),
        legend.text = rownames(quantity_matrix),
        args.legend = list(x = "topright"),
        ylim = c(0, 350))
```

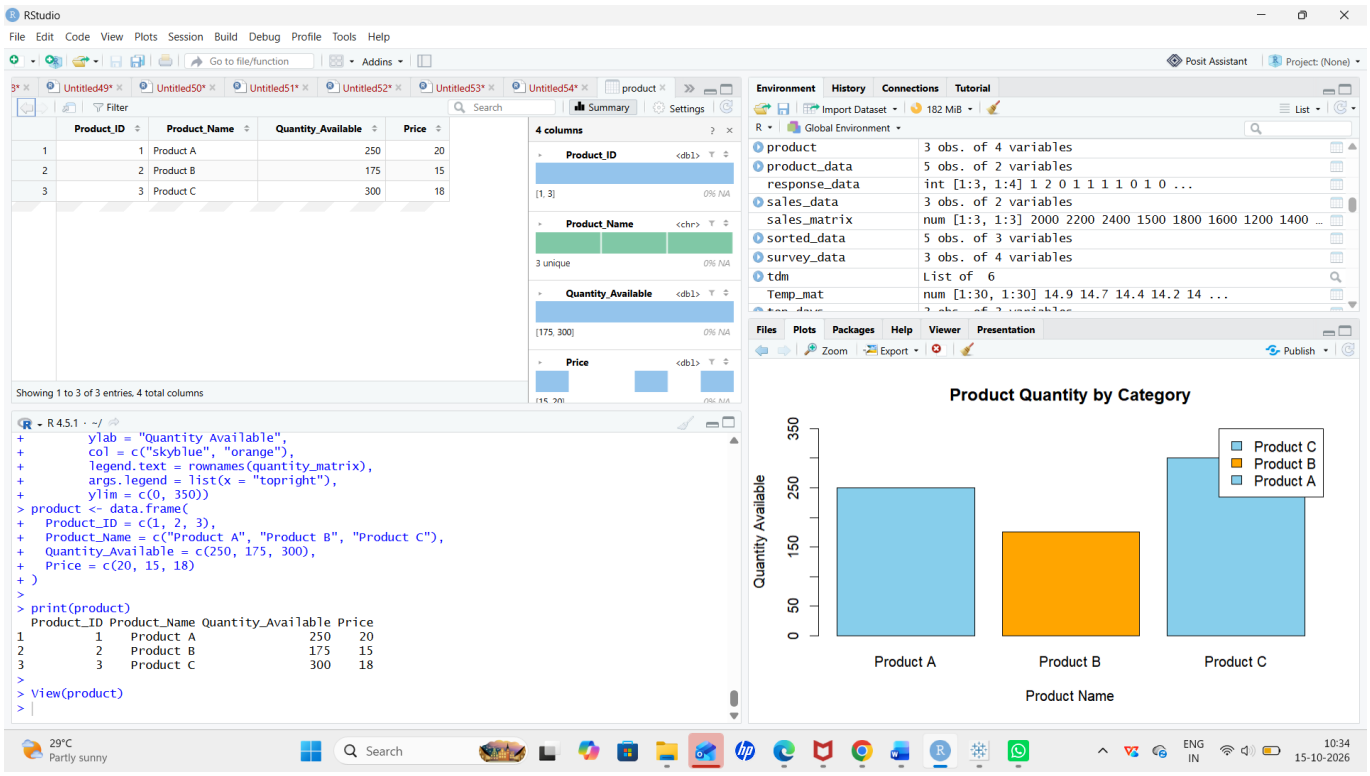


3. Build a table to display the performance data for each employee. Label the table elements.

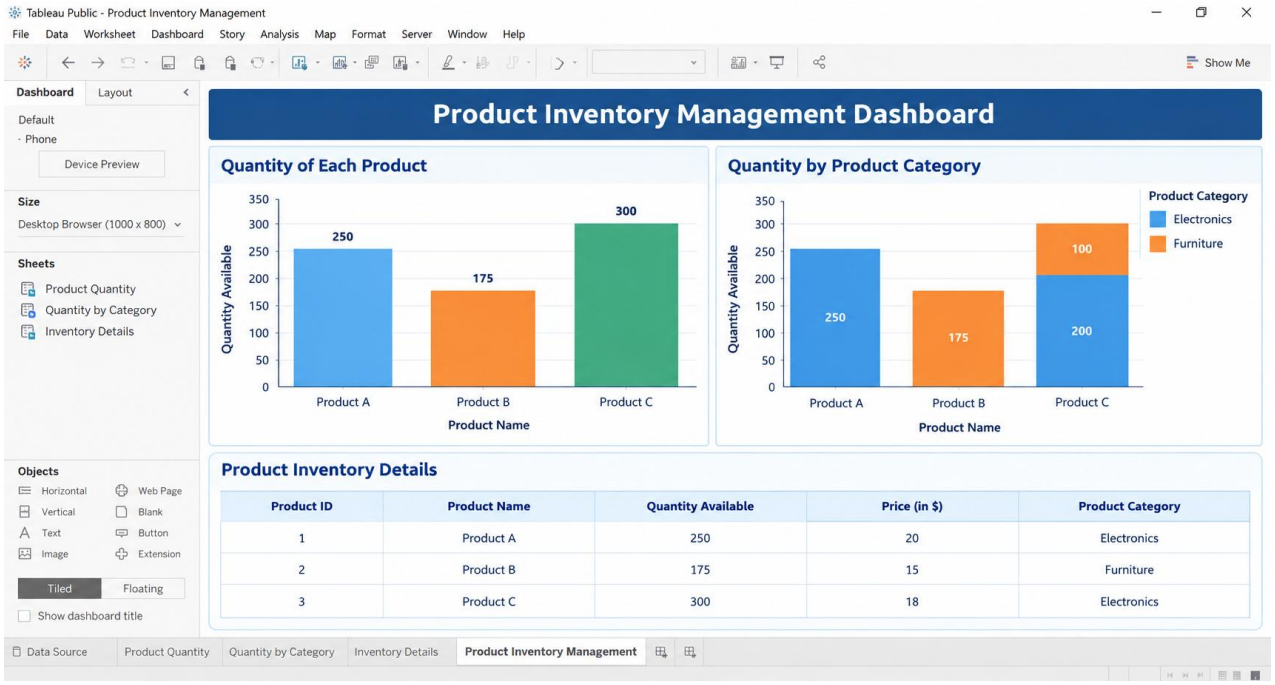
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product <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

print(product)

View(product)
```



4.Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive



Procedure

1. Prepare the product inventory dataset containing Product ID, Product Name, Quantity Available, and Price.

2. Open Tableau and connect the dataset using the Text File option.

3. Create a bar chart by placing Product Name on Columns and Quantity Available on Rows. Add axis labels and a suitable chart title.

4. Create a stacked bar chart by placing Product Name and Quantity Available on the appropriate shelves and using Product Category to divide the bars by category. Since the given dataset does not contain a Product Category column, add this column to the dataset before creating the chart.

5. Create a table displaying Product ID, Product Name, Quantity Available, and Price.

6. Create a new dashboard and combine the bar chart, stacked bar chart, and inventory table.

7. Adjust the layout, add a dashboard title, and save the Tableau workbook.

Result

Thus, the product inventory data was visualized using Tableau. A bar chart was created to compare product quantities, a stacked bar chart was designed to show quantities by product category, and a table was created to display inventory details. Finally, all visualizations were combined into a dashboard for inventory analysis.